

University of Yaoundé I
Department of Physics, Faculty of Science
P.O. Box 812, Yaoundé, Cameroon
myke-vital.sao@facsciences-uy1.cm
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The Editors
Digital Discovery
Royal Society of Chemistry

Dear Editors,
17 September 2026

We submit the manuscript entitled “**Chemical distribution shift in antimalarial natural products: fingerprint dominance and localized topological complementarity in molecular representation learning**” for consideration as a Full Paper in *Digital Discovery*.

This study speaks directly to your journal’s data-first mission: rather than another passive benchmark reporting fingerprint dominance, we deliver a distance-resolved characterisation of where classical fingerprint baselines and learned representations diverge under chemical distribution shift. On a frozen panel of 19 836 African antimalarial natural products and synthetic derivatives, evaluated across 25 fold–seed replicates per model, we establish three contributions:

First, we localise the ECFP4–RF advantage: nearest-neighbour Tanimoto decile stratification shows the largest margins in the structural extrapolation tail (similarity ≤ 0.40), narrowing sharply — without reversing — in mid-range interpolation. Across structural cohorts, fusing persistent-homology topological features (GIN–TFP) yields a small, replication-consistent gain over base GIN, recovering 36 % of the aggregate scaffold-split deficit; this localized topological complementarity is reported descriptively and is compatible with, but does not isolate, a 1-Weisfeiler–Lehman expressivity limitation of message-passing networks.

Second, we propose a Distance-Aware Conformal Triage Filter as a testable operational hypothesis: its post-hoc split-conformal coverage meets the nominal 90 % target on the five routed arms, and isotonic recalibration cuts residual expected calibration error 3 to 5-fold. We state explicitly that no prospective virtual-screening precision gain is claimed; prospective validation on an independent low-prevalence library is reserved for future work.

Third, we show that fold-independent weight restoration in molecular transformers reduces the risk of cross-fold information transfer during cross-validation — a reproducibility safeguard we recommend for transformer benchmarks.

Data and code: all splits, predictions, and analysis scripts are archived on Zenodo (DOI 10.5281/zenodo.19608875, record reserved pending upload completion) and GitHub (https://github.com/NanaEngo/Malaria_codesV2). Editors and referees will be given full access for peer review, and the DOI will be published before acceptance, per your data policy.

The work is original, is not under consideration elsewhere, has been approved by all authors, and there are no conflicts to declare. A TOC graphic ([figures/graphical_abstract.jpg](#)) accompanies the submission.

Suggested reviewers:

1. **Prof. Kelin Xia** (Nanyang Technological University, Singapore) — Topological deep learning, persistent homology, molecular graph representations. Email: xiakelin@ntu.edu.sg
2. **Prof. Fidele Ntie-Kang** (University of Buea, Cameroon / Martin Luther University Halle-Wittenberg, Germany) — African Natural Products Database (ANPDB), natural product cheminformatics. Email: fidele.ntie-kang@uni-halle.de
3. **Prof. Carsten Eickhoff** (University of Tübingen, Germany) — Molecular representation benchmarking, out-of-distribution generalisation. Email: c.eickhoff@uni-tuebingen.de

We thank you for your consideration.

Myke Vital Sao Temgoua
Corresponding Author